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<sol_start id=1>

1. (3) Concave mirror

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<sol_start id=2>

2. $u = 10 \text{ cm}$

$$m = -3$$

$$m = \frac{v}{u}$$

$$-3 = \frac{v}{-10}$$

$$v = 30 \text{ cm}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{r}$$

$$\frac{1}{r} = \frac{1+3}{30}$$

$$\frac{1}{r} = \frac{4}{30}$$

$$r = \frac{4}{30}$$

$$r = 7.5 \text{ cm}$$

(3) 30 cm, 7.5 cm

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3. (4) Both 2 and 3

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4. (i) Focus :- After reflection where the points converge I a concave mirror and appear to diverge from in a convex mirror to form an image is called the focus or principal focus as it lies on the principal axis, It is denoted by f.

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- (ii) Centre of curvature :- The centre of curvature is the centre of the sphere of which the spherical mirror is a part.
It is denoted by 'c'.

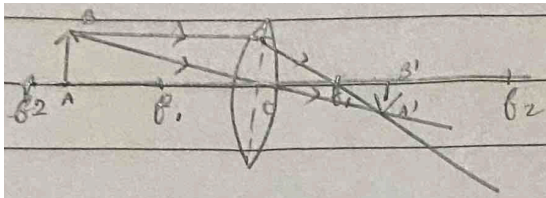
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5.

$$v = +18$$

$$m = \frac{h_i}{h_o} = \frac{v}{u} = \frac{-1}{5}$$

$$\frac{18}{u} = \frac{-1}{5}$$

$$18 \text{ rs} = -u$$

$$u = -90$$

$$\frac{1}{r} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{r} = \frac{1}{18} + \frac{1}{90}$$

$$\frac{1}{r} = \frac{5+1}{90}$$

$$r = \frac{90}{6}$$

$$r = 15 \text{ cm}$$

The nature of the lens is a convex lens, converging lens.

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6. (i) $h_o = 7 \text{ cm}$

$$u = -27 \text{ cm}$$

$$f = -18 \text{ cm}$$

By mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{r}$$

$$\frac{1}{v} = \frac{1}{r} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{-18} - \frac{1}{-27}$$

$$\frac{1}{v} = \frac{-3+2}{54}$$

$$\frac{1}{v} = \frac{-1}{54}$$

$$v = -54 \text{ cm}$$

$$m = \frac{h_i}{h_o} = \frac{-v}{u}$$

$$\frac{h_i}{7} = \frac{-(-54)}{-27}$$

$$h_i = \frac{\cancel{54} \times 7}{\cancel{27}}$$

$$h_i = -14$$

$$m = \frac{-14}{7}$$

$$m = -2$$

∴ The image is magnified and it's a real image and the screen should be placed 54 cm in front of the mirror to get the image.

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(ii) $R = 5 \text{ cm}$

$$u = -20 \text{ cm}$$

$$f = \frac{R}{2} = \frac{5}{2} = 2.5 \text{ cm}$$

$$\frac{1}{r} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{r} - \frac{1}{u} = \frac{1}{v}$$

$$\frac{1}{v} = \frac{1}{2.5} + \frac{1}{20}$$

$$\frac{1}{v} = \frac{10}{25} + \frac{1}{20}$$

$$\frac{1}{v} = \frac{40+5}{100}$$

$$v = \frac{100}{45}$$

$$v = 2.22 \text{ cm}$$

$$m = \frac{h_i}{h_o} = \frac{-v}{u}$$

$$= \frac{-2.22}{20}$$

$$= 0.11$$

$\therefore$ The image is smaller than the object, the image will be smaller, and the nature of image is virtual, upright and diminished.

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- (iii) The mirror used by the dentist is concave mirror. This is because concave mirror forms a virtual, erect and enlarged image when the object is placed within focus.

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